

Generalize ElGamal public key cryptosystem

$(G, *) \rightarrow$ abelian group

Discrete log problem on $(G, *)$

$(G, \alpha, \beta) \quad \alpha \in G, \quad \beta \in H$

where $H = \{ \alpha^{x^0} \mid x^0 \geq 0 \}$

$\downarrow$
subgroup of G
generated by α .

$\swarrow$
cyclic
gp

1. Select int a

2. Calculate $\alpha^a = \alpha * \alpha \dots * \alpha = \beta$

Find a ?

$(G, *) \rightarrow$ cyclic gp.

$\alpha \rightarrow$ generator or primitive element

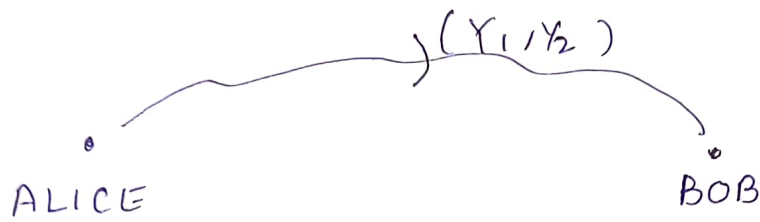
$$0 \leq a \leq |G| - 2$$

$$\beta = \alpha^a = \underbrace{\alpha * \alpha * \dots * \alpha}_{a \text{ times}}$$

Given (G, α, β)

Finding a is hard.

Plaintext space = G
 Ciphertext space = $G \times G$
 keyspace $K = \{(G, \alpha, a, \beta) \mid \beta = \alpha^a\}$ ^{secret}



ALICE

$$e_B = (G, \alpha, \beta)$$

Encryption

choose a k

$$E_{e_B}(x, k) = (Y_1, Y_2)$$

$$Y_1 = \alpha^k = \alpha * \alpha * \dots * \alpha \text{ (k times)}$$

$$Y_2 = x \cdot \alpha^k$$

BOB

$$(e_B, d_B)$$

Decryption

$$(Y_1, Y_2)$$

$$Y_2 * (Y_1^a)^{-1}$$

$$= x$$

$$k, 0 < k < G-2$$

Set up

1. $(G, *)$
2. α is generator
3. 'a' $\rightarrow$ Bob (private key)
4. $\beta \rightarrow \alpha^a$
 $\rightarrow \alpha * \alpha * \dots * \alpha$
 $\rightarrow$ public

ELGAMAL CRYPTOSYSTEM

based on discrete log problem (Diffie Hellman)

Discrete log problem over $\mathbb{Z}_p^*$

let p be a prime number then -

$$\mathbb{Z}_p^* = \mathbb{Z}_p - \{0\} \left[\begin{array}{l} \rightarrow \text{cyclic group} \end{array} \right]$$

let α be a generator (primitive element) of $\mathbb{Z}_p^*$

$$\beta \equiv \alpha^a \pmod{p}$$

where $0 \leq a \leq p-2$

Given: (β, α, p)

we need to find a ? $\left[\begin{array}{l} \text{DISCRETE LOG} \\ \text{PROBLEM} \end{array} \right]$

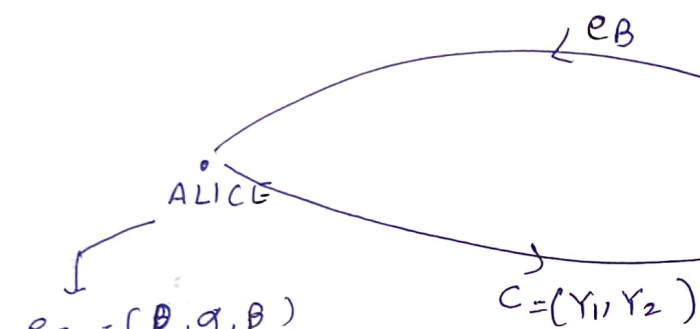
* If p is large then discrete log problem over $\mathbb{Z}_p^*$ is hard.

Time complexity is of the order of prime no. p .

→ BOB choose a large prime p such that discrete log prob is hard over $\mathbb{Z}_p^*$.

Elgamal:

public key set up a is a
secret key.



$$e_B = (p, \alpha, \beta)$$

encryption

$P = \mathbb{Z}_p^*$ (plaintext space)

$C = \mathbb{Z}_p^* \times \mathbb{Z}_p^*$ (C.t space)

choose a random number

$$k \in \mathbb{Z}_{p-1}$$

$$E_{e_B}(x, k) = (Y_1, Y_2)$$

$$\begin{aligned} Y_1 &= \alpha^k \mod p \\ Y_2 &= x \cdot \beta^k \mod p \end{aligned}$$

Key space

$$K = \{(p, \alpha, a, \beta)\}$$

$$\beta \equiv \alpha^a \mod p$$

$$K = (p, \alpha, a, \beta)$$

secret key

$$e_B = (p, \alpha, \beta)$$

$$d_B = a$$

private key

public key

$$Y_1 = \alpha^k \mod p$$

$$Y_1^a = \alpha^{ak} \mod p$$

$$= \beta^k \mod p$$

$$(Y_1^a)^{-1} = (\beta^k)^{-1} \mod p$$

$$= Y_2 \cdot (Y_1^a)^{-1} \mod p$$

$$= x \beta^k \cdot \beta^{k^{-1}} \mod p$$

$$= \boxed{x} \rightarrow$$

Set up BOB

1. choose a large prime p such that d.i.l. p is hard in $\mathbb{Z}_p^*$.

2. $\mathbb{Z}_p^* = \mathbb{Z}_p - \{0\}$
cyclic gp.

3. let α be the generator.

4. choose a secret a s.t.
 $0 < a < p-1$

$$5. \beta = \alpha^a \mod p$$

Example

$$p = 2579 \text{ (public)}$$

$$2. \alpha = 2 \text{ (public)}$$

$$3. a = 765 \text{ (private) key}$$

$$4. \beta = a^{765} \bmod p \\ = 949 \text{ (public)}$$

Encryption (ALICE)

$$x = 1299$$

$k = 853$ (random) chosen by ALICE

$$E_{EB}(x, k) = Y_1, Y_2$$

$$Y_1 = 2^{853} \bmod p = 435$$

$$Y_2 = 1299 \cdot 949^{853} \bmod 2579 \\ (x \cdot \beta^k \bmod p) \\ = 2396$$

Decryption

$$Y_1 = \alpha^k \bmod p$$

$$Y_1^a = \alpha^{ka} \bmod p$$

$$= \beta^k \bmod p$$

$$= 949^{853} \bmod 2579$$

$$Y_2 (Y_1^a)^{-1} = 2396 (435)^{-1} \bmod 2579$$

$$= 1299 \quad \checkmark$$